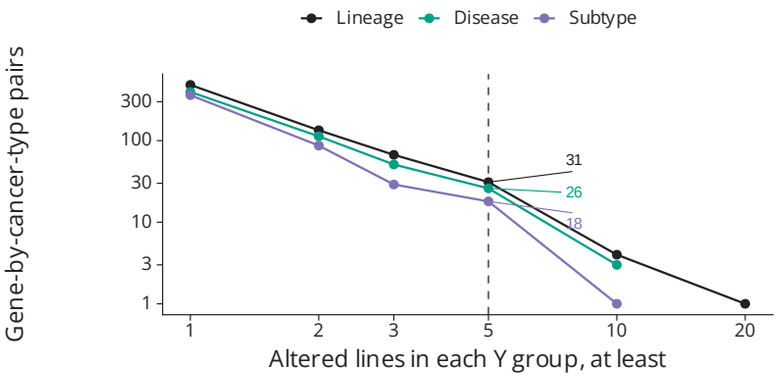
a

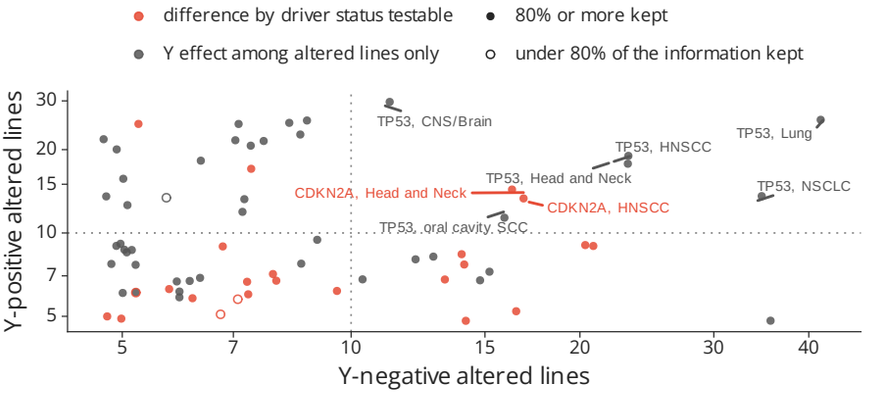
Of 2,869 gene-by-lineage pairs with an altered line, 31 have five altered lines per Y group



568 male CRISPR lines. A pair is a gene and a cancer type with at least one altered line (driver mutation, amplification or deep deletion). Dashed: the gate of five lines; the numbers are the pairs that reach it. A test of whether the Y effect differs by driver status also needs five unaltered lines per Y group: 13 pairs at lineage level, 8 at disease and 3 at subtype level.

b

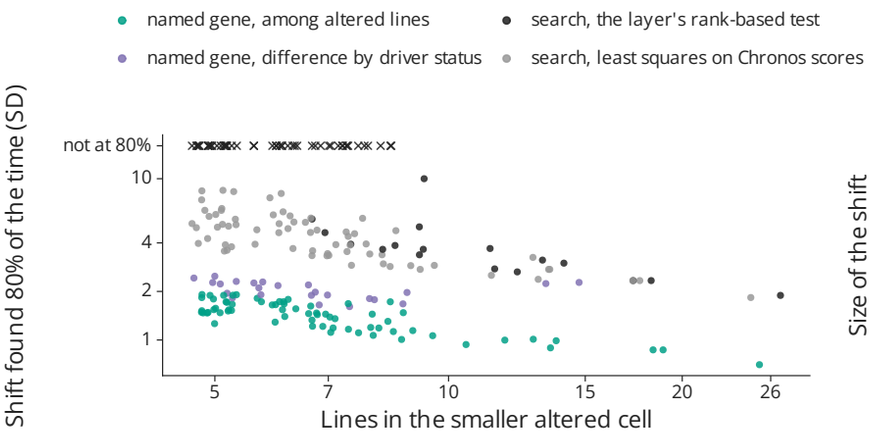
60 of the 68 testable splits have fewer than ten lines in the smaller altered cell



One point per split: a driver gene in a cancer type with five altered lines per Y group (68 splits in 22 cancer types and 14 genes; a split with the same lines at two levels is drawn once). Named: the 8 splits with ten lines in both cells. Open: screening library takes more than a fifth of the information (4 splits). Dotted: ten lines.

c

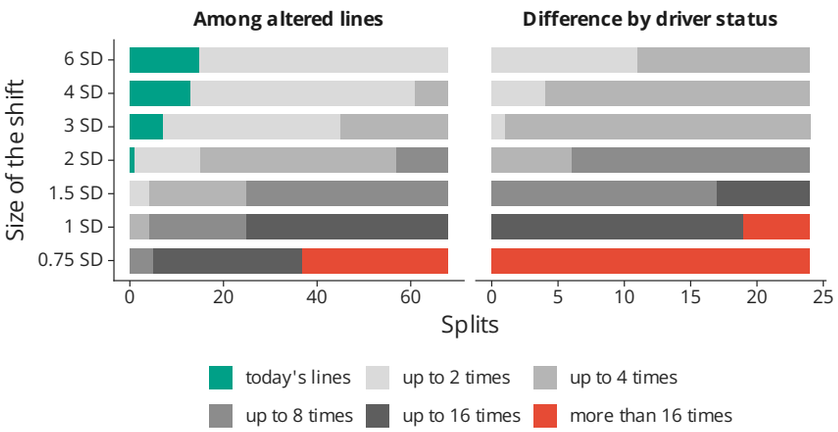
A gene named in advance needs 1.5 SD in the median split; a search needs 3.6 SD where the rank-based test reaches 80%, which is 17 of 68 splits



Shift in the Y-negative altered lines of a split, in SD of the gene's spread between lines, simulated on 500 real genes per shift, each tested as the one gene added to the split's fit. Named gene: $p < 0.05$. Search: the bar of the whole layer ($p < 1.2e-07$). Crosses: not found 80% of the time at any shift up to 10 SD; the rank-based test then finds a 10 SD shift in 11% to 79% of simulations, because ranks cap how far a few lines can move. Least squares on Chronos scores has no such cap and is not calibrated in cells this small; it is drawn for comparison. The difference by driver status can be tested in 24 splits; in a search it reaches 80% in none of them.

d

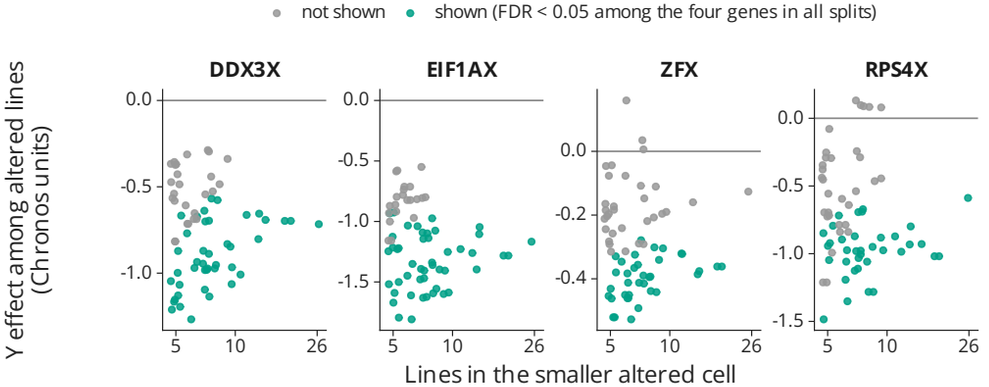
In a search, a shift of 2 SD needs 3.3 times today's lines in the median split and one of 1 SD 9.9 times; 1 of 68 splits finds a 2 SD shift today



For each split and each size: the multiple of today's lines at which the shift is found 80% of the time at the bar of the whole layer, with the layer's rank-based test, from simulations on 2, 4, 8 and 16 copies of the split's design (200 genes per shift; each copy carries the spread of a different real gene). Left: the Y effect among altered lines; right: the rule that also needs the difference from unaltered lines. The bar and the variance prior are today's. Checked on real splits cut down and copied back, the copies' limit at this bar is 6% lower than the real one on average, so the multiples are if anything somewhat low. For scale, the four paired genes' effects are 0.9 to 1.9 SD of their own spread and 1.5 to 6.9 SD of a typical gene's.

e

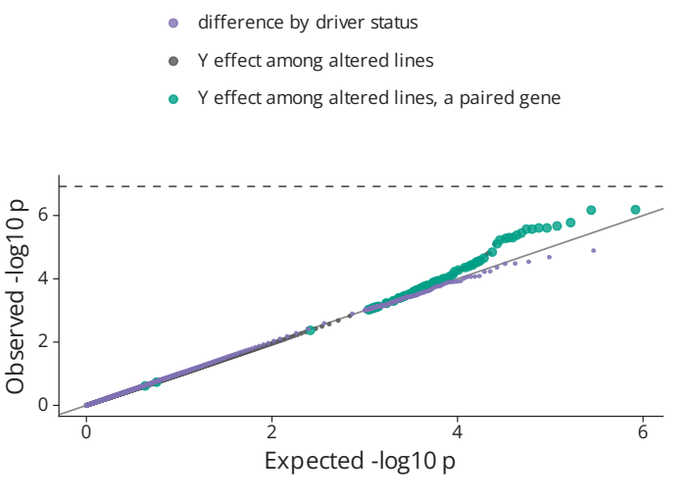
The four known Y-loss dependencies are seen in 161 of 272 tests inside splits: 40% with five lines in the smaller cell, 94% with ten or more



The four genes are named in advance, so they are judged as a family of their own (four genes in every split). A point is one gene in one split; the estimate is on the raw Chronos scale, the test on ranks.

f

No test passes; 12 of the 14 smallest distinct p-values belong to the four paired genes



414,052 tests of the Y effect among altered lines and 146,136 of its difference from the unaltered lines. Dashed: the p-value one discovery among altered lines would need (1.2e-07). Smallest: p below 1e-5, a test that two splits share counted once. With Y status shuffled within driver status, 12 of 200 runs gave a hit among altered lines and 0 a call.